
Analysis of mode choice behaviour in the London metropolitan area

Course assignment - 16 September 2026

1 Aim and content

The goal of this assignment is to develop a discrete choice model for analyzing the transportation mode choice behaviour of individuals living in the London metropolitan area. In this assignment, you will:

1. test different model specifications to obtain the best model possible,
2. apply the resulting model to forecast the changes in demand in response to changes in the attributes of the alternatives.

The description of the case study can be found on Moodle under the *Datasets* link (London passenger mode choice, LPMC). A sample of the original LPMC data is available in the assignment section of Moodle. All alternatives are available to all individuals.

2 Technical information

- All the concepts necessary to answer the questions are covered during the lectures and/or laboratory sessions.
- The results of your analysis must be documented in a report, **written in English**.
- Justify all your answers by explaining how the results were obtained and by reporting any formula used. Make sure answers are concise.
- All members of the group must contribute equally to all tasks!

3 Submission and grading

- A report must be submitted as a .pdf file on Moodle under *Assignments*. Please submit only the report. If I need any additional files (e.g., code associated with your model specifications), I will explicitly ask you to submit them.
- The deadline for submitting the report is **6 November at noon**.
- The assignment accounts for **20%** of your final grade. **20 points** will be distributed among the tasks described in Section 4.

4 Tasks

Model 0 [1.5 points] Develop a model specification that includes alternative specific constant, cost and travel time for each alternative. Cost and travel time are associated with generic parameters. Present both the specification (i.e., the utility functions) and the estimation results (parameter values, t-tests or p-values, null and final log-likelihoods). **[1 point]**

1. Comment on the estimation results (statistical significance and sign of all parameters). **[0.5 point]**

Model 1 [3 points] Using Model 0 as the base model, include alternative-specific parameters for one of the attributes of Model 0. Present both the specification and the estimation results (as defined previously). **[1 point]**

1. State the underlying assumption of defining alternative-specific parameters in this specific situation. **[0.5 point]**
2. Comment on the estimation results (as defined previously, including any changes from the previous model). **[0.5 point]**
3. Compare Model 0 and Model 1 with an appropriate statistical test. Explain your choice of test. State the null hypothesis and the result of the test. Denote the preferred model as $\text{Model}_{\text{pref}}$. **[1 point]**

Model 2 [4 points] Using $\text{Model}_{\text{pref}}$ as the base model, include one additional alternative attribute and one interaction of a socioeconomic characteristic with either the ASCs or one of the attributes. Present both the specification and the estimation results (as defined previously). **[1 point]**

1. State the underlying assumptions of the additional attribute and interaction in this specific situation. **[1 point]**
2. Comment on the estimation results (as defined previously). **[1 point]**
3. Compare $\text{Model}_{\text{pref}}$ and Model 2 with an appropriate statistical test. Explain your choice of test. State the null hypothesis and the result of the test. Denote the preferred model as $\text{Model}_{\text{pref}}$. **[1 point]**

Model 3 [3 points] Using $\text{Model}_{\text{pref}}$ as the base model, include an appropriate non-linear transformation of one of the variables. Present both the specification and the estimation results (as defined previously). **[1 point]**

1. State the underlying assumption of the non-linear specification defined in this situation. **[0.5 point]**
2. Comment on the estimation results (as defined previously). **[0.5 point]**
3. Compare $\text{Model}_{\text{pref}}$ and Model 3 with an appropriate statistical test. Explain your choice of test. State the null hypothesis and the result of the test. Denote the preferred model as $\text{Model}_{\text{pref}}$. **[1 point]**

Market shares [2.5 points] Assume that stratified random sampling was used to produce your sample. The sample was divided into strata (i.e., groups) and simple random sampling was used to sample observations in each stratum. We consider the following strata:

- **S1:** females aged 40 years or younger;
- **S2:** females aged 41 years or older;
- **S3:** males aged 40 years or younger;
- **S4:** males aged 41 years or older.

Table 1 gives the size of each category in the full population.

	Age ≤ 40	Age > 40
Male	2'676'249	1'633'263
Female	2'599'058	1'765'143

Table 1: London population estimates in 2015 (Source: ONS)

1. Report the size and weight of each stratum in your sample [**1 point**].
2. Using $\text{Model}_{\text{pref}}$ and the weights of your strata, compute the predicted market share of each mode and their confidence intervals. Do the obtained results match your expectations? [**1 point**]
3. Compare the market shares predicted by $\text{Model}_{\text{pref}}$ with the weighted market shares computed using the actual choices. [**0.5 point**]

Forecasting [6 points] Consider the following scenarios: (i) an increase of car cost by 15%; and (ii) a decrease of public transport cost by 15%.

1. Report the market shares predicted by $\text{Model}_{\text{pref}}$ for each scenario. Do they match your expectations? Compare the predicted market shares with the original market shares. [**1 point**]
2. Which scenario is the most effective policy if the goal is to decrease the share of car? Explain why. [**0.5 point**]
3. Calculate the total public transportation revenue in each scenario. Which scenario reports the highest total revenue? Explain why. Is it higher than the total revenue obtained without any of the policies? Can you explain why? [**1.5 points**]
4. Calculate the average value of time for car and public transportation. Comment on the obtained results. [**1 point**]
5. Compute the direct and cross aggregate elasticities of car cost and public transport cost. Comment on the obtained results. Report the normalization factors. [**2 points**]